20 * Breadth first search.

Algorithm:-

i) enter the starting node in the queue.

ii) If queue is empty, return fail and stop

iii) If queue first element is goal node, return successful and STOP.

iv) else, Remove ~~the~~ and expand the first element in queue and place child nodes at the end of the queue.

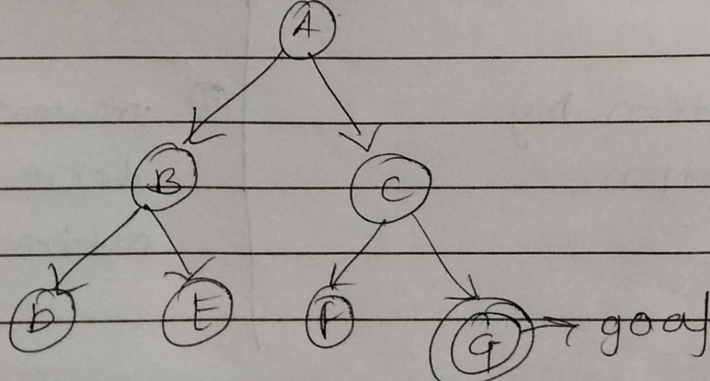
v) Go to step ii) until the goal node is found.

① Breadth first search is an uninformed search algorithm.

② It explores all the nodes level by level in the search tree.

③ It expands the all nodes at present depth before moving to next depth level.

④ It uses a queue and FIFO (first in first out)



{A} → ~~A~~

{B, C} → ~~B~~

{C, D, E} → ~~C~~

{D, E, F, G} → ~~D~~, ~~E~~, ~~F~~.

Breadth First Search (BFS)

Subject: Artificial Intelligence

1️⃣ Explanation of BFS (5 Points)

1. Breadth First Search (BFS) is a search technique used to explore nodes level by level.
2. It explores all nodes at the present depth before moving to the next level.
3. It follows the rule **"Go wide first, then deep."**
4. It uses a **queue (FIFO – First In First Out)** data structure.
5. It is an **uninformed search algorithm**.

2️⃣ Algorithm of BFS (5 Steps)

1. Start from the root (initial) node.
2. Insert the root node into a queue.
3. Remove the front node from the queue and check if it is the goal.
4. If not, insert all its unvisited child nodes into the queue.
5. Repeat until the goal is found or the queue becomes empty.

3️⃣ Working Principle of BFS (5 Points)

1. BFS explores nodes level by level.
2. It visits all neighbors before going to the next depth level.
3. It always explores the earliest inserted node first.
4. It continues until the goal node is found.
5. If all nodes are visited and goal is not found, search fails.

4️⃣ Characteristics of BFS (5 Points)

1. Uses a **queue (FIFO)**.
2. Guarantees the **shortest path** in unweighted graphs.
3. Requires more memory than DFS.
4. Time complexity: **O(b^d)** (b = branching factor, d = depth).
5. Suitable when solution is near the root.

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Q) Explain Depth first Search (DFS). Describe its algorithm, working principle & characteristics.

→

• Description :

Depth first Search is an Algo. that starts ~~with~~ at the root and explores as far as possible along each branch (path) before backtracking. It goes "deep" into tree first.

" Go deep first then backtrack "

• Algorithm :

1) Start from initial (root node) node.
2) Push the node into stack.
3) Pop the top node & check if it is goal.
4) If not, push the unvisited child nodes into stack.
5) Repeat until the goal is found or stack becomes empty.

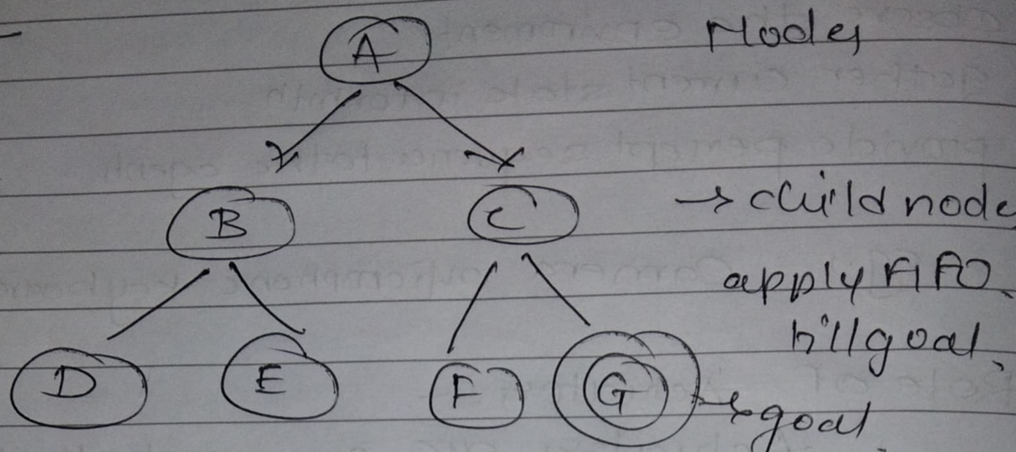
• Working principle :

1) DFS goes as deep as possible in one direction
2) When no further nodes are available, it backtracks
3) It always explores the most recently added nodes first.
4) It continues this process until the goal is found
5) If all nodes are visited and goal is not found, search fails.

• Characteristics :

1) Uses stack or recursion
2) Memory requirement is less compared to BFS.
3) Time Complexity $O(b^m)$
4) Does not guarantee shortest path.

BFS



{A} → A ×

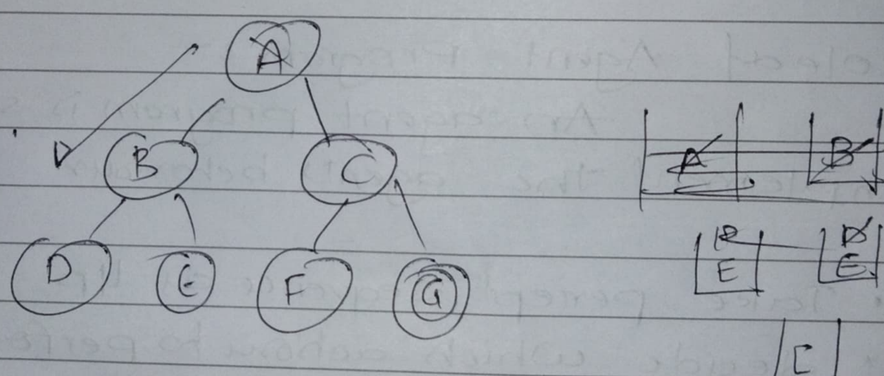
{B, C} → B ×

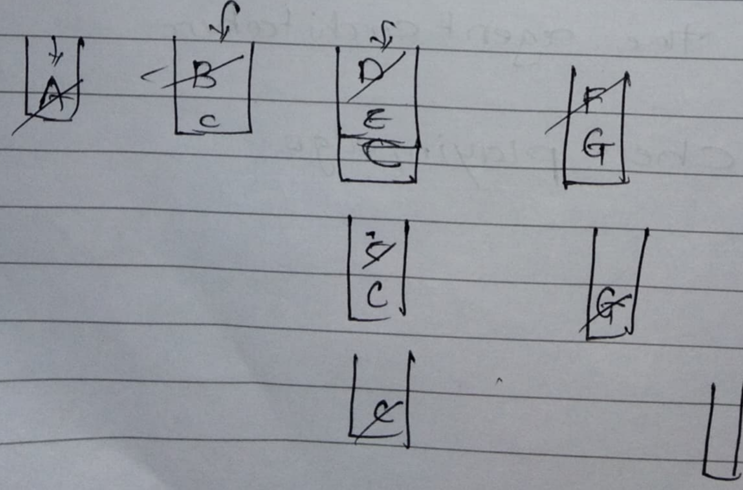
{C, D, E} → C ×

{D, E, F, G} → D E F × × ×

{G} → goal

DFS





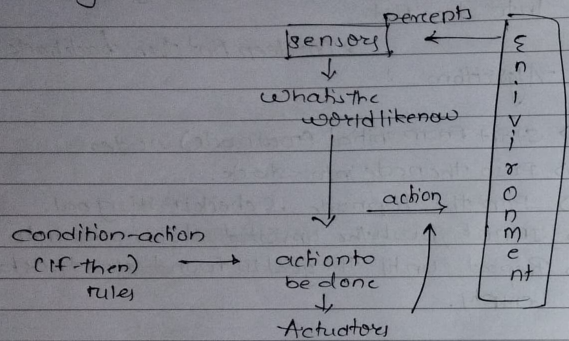
Q7 Compare: BFS & DFS.

Feature	BFS	DFS
Abbreviation	Breadth first search	Depth first search
Strategy	Explores level-by-level (Breadthwise)	Explores path-by-path. (depthwise)
Data str.	Uses a queue (FIFO)	Uses a stack (LIFO).
Completeness	Complete: Always finds a soln if one exists	Incomplete: May get stuck in ∞ loops.
Optimality	Optimal [Always finds shortest path]	Not optimal [May find a longer path]
Space Complexity	High [Needs lots of memory to store all nodes at current level]	Low [Needs less memory to store just the current path]
Application	GPS	Puzzles/mazes

Q] Explain the structure of an Intelligent agent. Describe role of sensors, actuators and agent programs.

→

An Intelligent agent is defined as agent is an is an entity that perceives its environment through sensors and acts upon that environment through actuators to achieve its goals.



- Components :-

1. Environment
2. Sensors
3. Agent
4. Actuators
5. Agent program

- Role of Sensors :-

Sensors are device that collect informtn from the Environment & convert it into percept for agent.

- Observe the environment
- gather current state informtn
- provide percept sequence to the agent.

eg., Camera Microphone keyboard.

- Role of Actuators :-

Actuators are component that execute actions chosen by agent in environment.

- Affect the environment
- performs physical or digital actions
  eg., Motors, Wheels, Robotic arms
- Screen display

- Role of Agent Program :-

An agent program is s/w that implement the agents behaviour

- Take percept sequence as I/P.
- decide which actions to perform
- Run on the agent architecture

eg., Chess playing algo.

9) Simple Reflex Agent

Sensor + current state of envir

Percept ← input

Environment

↓

Action should be taken

Condn Action Rule →

+ Actuators

Action →

· Limitation
1. very less Intelligent
2. Conged itself stuck in loop.

8) Explain the History and evolution of Artificial Intelligence. Highlight important milestones and contributions in the development of AI.

→

- Contributions :-

1) Alan Turing - Introduced Turing test (1950)
2) John McCarthy - Coined the term AI; developed Lisp programming language
3) Newell & Simon - Developed logic theorist
4) Marvin Minsky - Major contributor to AI research & networks
5) Geoffrey Hinton - father of Deep learning

- Evolution of AI :-

1) The Birth of AI (1950-1956):
   1950 : Alan turing "Turing test" to check if a Machine can think like human
   1956 : The term "AI" was coined by John McCarthy

2) The Golden Years (1956-1974):
   Researcher created program that could solve algebra & prove theorems.
   ELIZA (1966) : first chatbot was created

3) AI Winter (1974-1980):
   funding ~~for~~ AI stopped because the promises made by researchers (like "Machine will replace human in 10 years") were not fullfilled.

4) Boom of Expert Systems (1980-1987):
   AI cameback with "Expertsystem" which could make decision like human
   MYCIN (medical diagnosis system)

5) 2nd AI Winter (1980s - 1990s)
   Slowed down

6) Machine learning Era (1990s - Present):
   1997 : IBM's DeepBlue Computer beat the World CM, Champion Garry kasparov
   2010 : Deeplearning Introduced, & Bigdata
   eg., Siri, Alexa,

Q) Explain the Applications of AI in realworld scenarios. Also discuss current trends in AI.

→ • Appln:-

1. Transport & tourism (Website
2. Self Driving car (AI based car
3. E commerce (Amazon
4. In Gaming (chess AI
5. In Entertainment (Amazon Prime, Netflix

6. You musthave seen use of Artificial intelligence in many sci-fi movies. To name a few we have I Robot, Wall-E, The Matrix ~~Theory~~ Triology, Star Wars, etc movies.

- Many a times these movie show positive potential of using AI and sometimes also emphasize the dangers of using AI. Also there are game based on such movies. Which shows us many probable appln of AI.

6. In medical

(AI has appln in field of Cardiology (ECG), Neurology (MRI), etc complex operations of internal organs, It can be also used in organizing bed schedules, managing staffs rotation, store & retrieve info. for patient, etc.

7. In Military

Training stimulators can be used. AI is a natural partner in Modern Military.

• Current trends:-

- Artificial Intelligence has touched each and every aspect of our life.
- from Washing Machine, Air Conditioners, to smart phones everywhere AI is serving to ease our life.
- companies are using AI to improve their product and increase sales.
- AI saw significiant advances in ML. Following are the areas in which AI is showing significant advancements

a. Generative AI:

Tools like ChatGPT and Midjourney that can create new text, images, and code.

b. AI in IOT:

Some smart homes where device talk to each other to save energy.

c. AI replacing workers:

In industry where there are safety hazards, robot are doing a good job. Human resources are getting replaced by robots rapidly.

III] Neuroscience :

- The study of how the biological brain works. AI researchers study neurons in the human brains to create "Artificial Neural Networks".
- Neural networks inspired by the brain.

IV] Pyschology & Cognitive Science :

- Cognitive pyschology studies how humans think and learn. AI uses these theories to mimic human behaviour in machines.
- Human thinking & learning behavior.

V] Linguistics :

- The study of language. To understand and speak human languages (like Siri or Google Translates), AI needs the theories of grammar and syntax from linguistics (Natural Language Processing).
- Rules for Language understanding (NLP).

Q] Discuss the major sub-areas of Artificial Intelligence. Explain any 5 subareas briefly.

→ AI is a huge umbrella term that covers many specialized fields.

- Major Sub-areas :

I] Machine Learning (ML):

- The art of getting computers to act without being explicitly programmed. It focuses on algorithms that learn from data.

II] Natural Language Processing (NLP):

- Focuses on the interaction between computers and human language.
  Eg:, Spam filters in email, Chatbots, Languge translation

III] Computer Vision :

- Enables computers to "see" and interpret images or videos.
- Example:, Face recognition on your phone, Medical X-ray analysis.

IV] Robotics :

- Focuses on creating physical robots that can move and interact with the real world.
  Eg:, Roomba vacum cleaners, Mars Rovers.

V] Expert Systems :

- Computer programs that solve complex problems by reasoning through bodies of knowledge represented mainly as if-then rules.
  Eg, flight tracking systems, Clinical support systems.

11 Feb 2026

* Answers for UTI QB :—

Q] What is an Intelligent system? Explain the categorization of intelligent Systems with suitable examples?

→

• Definition:

- An Intelligent System is a machine with a embedded, Internet-connected computer that has the capacity to gather and analyze data and communicate with other systems.
- It take input from the world, process it and produces an output.

• Categorization of Intelligent System:

I] Rule-Based System:

- These systems work on pre-defined rules set by human experts.
- They follow "If-Then" Logic.
  Eg., A simple medical diagnosis system (If fever >100 and cough, then check for flu).

II] Learning-Based System (Machine Learning):

- These systems do not need hard-coded rules. Instead, they learn patterns from data.
- The more data they see, the better they get.
  Eg., Image based analysis, Netflix recommendation system (It learns what movies you like based on what you watched)

III] Hybrid Systems:

- These system combine both Ruled-based + learning-based approaches to solve examples.
  Eg., Autonomous vehicles using rules and ML for navigation

Q] Explain the foundations of Artificial Intelligence. Discuss any 5 disciplines that contribute to AI.

→

- AI is not a standalone subject. It is built upon a knowledge of multiple disciplines:

• contributing disciplines:

I] Philosophy:

- Provided a initial ideas about the mind. Philosophers asked: "Can a machine think?" and "Where does knowledge come from?" It gave the AI the concepts of logic and reasoning.
- Ethics, logic, mind & conciousness debates.

II] Mathematics:

- Provided the tools to manipulate statements logically (Boolean algebra) and handle uncertainty (Probability theory). Without maths, we cannot write algorithms.
- Linear algebra, calculus, probability, statistics.